LaB 4, Day 1 Ian Havill

1) $z_1^{(1)} = w_{11}x_1 + w_{12}x_2 + b_1^{(1)}$

$= (0.5)(1) + (-0.3)(2) + 0.1$

$= \boxed{0}$

2) $h_1 = \text{RELU}(z_1^{(1)})$

$= \boxed{0}$

3) $z_2^{(1)} = w_{21}x_1 + w_{22}x_2 + b_2^{(1)}$

$= (0.8)(1) + (0.2)(2.0) + (-0.1)$

$= \boxed{1.1}$

4) $h_2 = \text{ReLU}(z_2^{(1)}) = 1.1$

5) $z^2 = v_1h_1 + v_2h_2 + b^{(2)} = (0.6)(0) + (-0.4)(1.1) + (0.2)$

$= -0.24$

6) $\hat{y} = \sigma(z^{(2)}) =$

$\sigma = \frac{1}{1+e^{-x}} = 0.44028$

7.) $L = -[y\log(\hat{y}) + (1-y)\log(1-\hat{y}))$

$-[-0.8203] = 0.82032$